

CERTIFIED REDUCED-LEAD ECG RECONSTRUCTION: WHAT IS RECOVERABLE, AND WHAT IS FABRICATED

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ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is an ill-posed inverse problem, and a scalar reconstruction error says nothing about *which* features a clinician can trust. We give a per-feature (P/QRS/ST/T), per-lead *recoverability certificate*. Its engine is a physical fact generic inverse-problem theory lacks: each waveform segment is an approximately rank-3 cardiac *dipole* plus a non-dipolar residual. This yields, *independent of any reconstructor*: (i) a closed-form conditioning $\kappa_s(S)$ that says which lead configurations can recover a feature’s dipolar content and how well—so six coplanar limb leads recover the transverse dipole far worse than three well-chosen leads; (ii) a distribution-free calibrated interval for the population-predictable residual; and (iii) a flag, with a *held-out* false-flag rate $\leq \alpha$, for content a reconstruction places in the certified-unrecoverable subspace. The certificate warns *upfront* that a configuration cannot support a decision—e.g. precordial ST from limb leads—which no per-reconstruction error reveals. Scoring a real conditional diffusion model against a held-out recoverability oracle, we show that optimising for realism injects a growing recoverability deficit—content the certificate proves recoverable yet the model increasingly trades for population-typical morphology—invisible to scalar error: fabrication is a property of the *objective*, and the certificate measures it.

Index Terms— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads. Dozens of recent methods—linear, U-Net, transformer, and diffusion—report low mean-squared error [1]. Yet a low aggregate error certifies nothing about *which* morphological features are trustworthy. Generative reconstructors optimise realism, and by the perception-distortion tradeoff [2] realism on an ill-posed inverse problem is bought by *inventing* content. On an ECG this is not a cosmetic artifact: a fabricated ST-segment deviation is a fabricated diagnosis.

A recent line of work formalises hallucination in generic inverse problems $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}$: [3] derives computable, forward-model-dependent bounds on hallucination magnitude and a test for the faithfulness of reconstructed detail, and [4] a calibration-free indicator of distribution shift—both for an *arbitrary* $\mathbf{A}$. The ECG is special: its forward operator carries a *physical* structure—a low-rank dipole and a physically meaningful non-dipolar null space—absent from a generic $\mathbf{A}$. This lets us make a *positive*, closed-form, *per-feature* statement (which named cardiac feature, on which lead, is recoverable and with what noise gain), name the unrecoverable content physically, and meter it clinically—a specialisation the forward-model-only bounds do not provide. Closest to us, [5] uses a U-Net to *quantify* each lead’s information content empirically; our certificate is closed-form, per-segment, and calibrated, derived from cardiac physics rather than a learned model.

Contributions. (1) A per-segment operator decomposition of reduced-lead ECG into three tiers—provably recoverable (dipolar), statistically recoverable, and provably unrecoverable—with a closed-form per-configuration conditioning $\kappa_s(S)$ that explains why 6 coplanar limb leads recover the (transverse) dipole far worse than 3 well-chosen leads, *before* and *independent of* any reconstructor (Sec. 2). (2) A certified estimator: group-conditional conformalised quantile regression for the recoverable non-dipolar part, and a conformal-risk-controlled hallucination flag, with a held-out false-flag guarantee, for the unrecoverable part (Sec. 3). (3) Evidence on PTB-XL that the certificate localises and *explains* reconstruction error where a scalar cannot: it flags configurations that cannot support a decision (precordial ST from limb leads) upfront, and—scoring a real conditional diffusion model against a held-out recoverability oracle—shows that fabrication is chosen by the *objective*, injecting a growing recoverability deficit a scalar error cannot see (Sec. 4).

2. THE RECOVERABILITY CERTIFICATE

Lead algebra. The standard 12-lead vector $\mathbf{L} \in \mathbb{R}^{12}$ has algebraic rank 8: the six limb leads are fixed linear combinations of {I, II} (Einthoven, Goldberger), so $\mathbf{L} = \mathbf{T}\mathbf{x}$ with a known $\mathbf{T} \in \mathbb{R}^{12 \times 8}$ and $\mathbf{x} = [\text{I, II, V1–V6}]$. Observing a lead subset S is exact row selection $\mathbf{y}_S = \text{Sel}_S \mathbf{L}$; the operator is

known, so we never estimate it.

Dipolar coordinates. Within a waveform segment $s \in \{P, QRS, ST, T\}$ the instantaneous potential is approximately a rank-3 cardiac dipole. We estimate a per-segment dipolar basis $M_s \in \mathbb{R}^{12 \times 3}$ (top-3 left singular vectors of the population, segment- s lead covariance) and mean μ_s , and write $L_s = \mu_s + M_s d + r_s$ with r_s the non-dipolar residual. We do *not* assume exact rank 3; M_s is a coordinate system and the residual is handled by calibration. On PTB-XL the dipolar fraction is 0.92 for QRS but only 0.74 (ST) and 0.58 (P): the recoverable share is genuinely feature-dependent.

Theorem 1 (Per-feature dipolar recoverability). *Let $M_{s,S} = \text{Sel}_S M_s \in \mathbb{R}^{|S| \times 3}$ and $\hat{L}_s = M_s M_{s,S}^+ y_S$. (i) If $\text{rank}(M_{s,S}) = 3$ the estimator recovers the population-dipolar projection with error $\hat{L}_s - \Pi_{\mathcal{D}_s} L_s = M_s M_{s,S}^+ (\text{Sel}_S r_s + n)$, hence $\|\hat{L}_s - \Pi_{\mathcal{D}_s} L_s\| \leq \kappa_s(S) (\|\text{Sel}_S r_s\| + \|n\|)$ with the closed-form gain $\kappa_s(S) = \|M_s M_{s,S}^+\|_2$; recovery is exact in the purely-dipolar limit $r_s = 0$, and otherwise carries a residual-contamination term amplified by the same $\kappa_s(S)$. (ii) If $\text{rank}(M_{s,S}) < 3$, the unobserved dipole directions are unrecoverable at any SNR. (iii) r_s splits into a y_S -predictable part (Tier II) and a y_S -independent part (Tier III).*

The proof is linear algebra; its value is that $\kappa_s(S)$ is a *per-configuration, per-segment* number a generic inverse-problem analysis cannot produce, because it has no M . It is honestly a *conditioning* statement: the same gain that bounds the noise also amplifies the observed non-dipolar residual, so a large $\kappa_s(S)$ warns that a feature is untrustworthy *whatever* the reconstructor. We report $\kappa_s(S)$ at full precision as this conditioning number and truncate directions below tolerance $\varrho=10^{-2}$ in the deployed estimator (treating them as Tier III), so e.g. the six limb leads have effective dipole rank 2 for the transverse component. Classical vectorcardiography [6] gives one whole-beat dipole map; here the subspace is per-segment.

The unrecoverable tier. On the y_S -independent part u (Tier III), $p(u \mid y_S) = p(u)$, so the Bayes estimate is the prior mean and $\mathbb{E}\|\hat{u} - u\|^2 \geq \text{Var}(u)$ for any estimator; any deviation from the mean strictly increases error. This is the standard non-identifiability limit [3, 4]; we *use* it, crediting the generic result, to justify the flag below.

Hallucination energy. With R_s the projector onto the recoverable dipole subspace and $U_s = I - R_s$, the per-lead $h_{s,\ell} = \text{RMS}_t [U_s(\hat{L}(t) - \mu_s)]_\ell$ is the energy a reconstruction places in the certified-unrecoverable subspace—a *deployable* scalar (no ground truth). It detects fabrication in the null space; error inside R_s (a contaminated dipole estimate, Thm. 1(i)) is instead warned by $\kappa_s(S)$. To *validate* that flagged energy is truly fabricated we also report an oracle diagnostic—the correlation ρ of $U_s(\hat{L} - \mu_s)$ with the true non-dipolar content—which needs the ground-truth leads and is not itself deployable. Geometry produces h ; calibration (next) turns it into a guaranteed flag.

3. DISTRIBUTION-FREE CALIBRATION

Tier II intervals. For each group $g = (s, \ell)$ we run conformalised quantile regression [7] with a separate conformal correction per group (Mondrian / group-conditional), so coverage holds conditional on the feature and lead. Given calibration quantiles (q_{lo}, q_{hi}) and scores $E_i = \max(q_{lo,i} - y_i, y_i - q_{hi,i})$, the corrected interval is $[q_{lo} - Q_g, q_{hi} + Q_g]$ with Q_g the finite-sample $(1 - \alpha)$ conformal quantile of $\{E_i\}_g$.

Tier III flag. We threshold h at τ_g , the one-sided $(1 - \alpha)$ conformal quantile of h over *faithful* reconstructions, so the false-flag rate is $\leq \alpha$, finite-sample and distribution-free; this absorbs the M_s estimation error automatically. Conformal risk control [8] sets α against the clinical loss (a missed STEMI $\gg$ a false alarm). Detection power is a measured curve with an SNR-set floor: we do not claim a two-sided guarantee.

Device shift. Under PTB-XL $\rightarrow$ CinC-2021 shift, Tier I exactness and Tier III *soundness* are distribution-free by construction (the null space is unobservable regardless of distribution); only Tier II *coverage level* is shift-sensitive. We measure the gap and close it with weighted conformal [9] plus small-slice recalibration.

4. EXPERIMENTS

Data. PTB-XL [10] (21,799 twelve-lead records, 100/500 Hz; recommended 10-fold split, folds 1–8 train, fold 10 test) with P/QRS/ST/T delineation by NeuroKit2 [11]; reduced-lead acquisition is the setting of the PhysioNet/CinC-2021 challenge [12], and we evaluate device shift across PTB-XL’s own acquisition devices (Schiller CS vs. AT families). Every theorem is cross-checked by simulation before use.

4.1. Synthetic validation

On controlled data with a known dipole and injected non-dipolar content of energy δ , all three claims hold exactly (Fig. 1). Tier I error grows linearly with noise at slope $\|M_s M_{s,S}^+\|_F$, and a collinear triplet ($\kappa=204$) is far worse than a spread one ($\kappa=15$). On Tier III the Bayes-optimal estimator’s error equals the $\text{Var}(u)$ lower bound at every δ , while a hallucinator that injects realistic content doubles the error. The flag’s false-flag rate is $0.084 \leq \alpha = 0.1$ with power reaching 1.0 for any non-trivial δ , and Mondrian-CQR attains 0.90 coverage.

4.2. Geometry, not lead count

On the PTB-XL QRS dipolar subspace, $\kappa_s(S)$ ranks configurations by *conditioning*: a single Lead I observes one of three dipole directions (effective rank 1); a spread triplet $\{I, II, V2\}$ gives $\kappa=3.1$ (rank 3); three adjacent precordials $\{V1, V2, V3\}$ give $\kappa=4.8$; but three *coplanar* limb leads

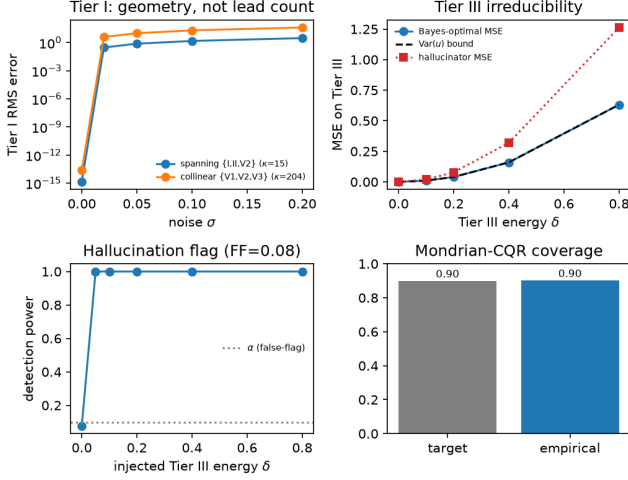


Fig. 1. Synthetic validation. Tier I noise gain (κ); Tier III irreducibility (Bayes = $\text{Var}(\mathbf{u})$; hallucinator $2\times$); flag power vs. false-flag rate; group-conditional coverage.

$\{I, II, III\}$ and the six limb leads have κ of order 10^5 and 10^4 and *effective rank 2*—their third (transverse) dipole direction is Tier III. This is a property of the lead set alone, available before any reconstruction: it says, upfront, that limb leads cannot recover the transverse-dipole content that precordial ST depends on.

4.3. The certificate localises and explains

Table 1 reconstructs the unobserved leads from $\{I, II, V2\}$ with three methods. A scalar RMSE ranks them (the generative model is worst) but does not say *which* feature is wrong or *why*. The certificate does. The dipolar baseline places *zero* energy in the certified-unrecoverable subspace ($h=0$): it never fabricates, but recovers only Tier I. The learned linear (OLS) reconstructor places moderate energy there whose oracle correlation ρ with the true non-dipolar content is *positive*—it genuinely recovers Tier II. The generative reconstructor places more energy there, uncorrelated with truth ($\rho \approx 0$). We stress the honest scope: ρ is an oracle validation, and h alone need not separate blur from fabrication when energies match—the deployable guarantee is the configuration-level certificate (κ , tiers) plus the held-out flag.

4.4. A real diffusion model fabricates recoverable morphology

The strongest test is non-circular: on the certified-unrecoverable subspace $\rho(\hat{\mathbf{L}}, \mathbf{L}) \approx 0$ for *any* sampler, so a low ρ there measures the definition, not fabrication. We therefore build a *held-out oracle*: a supervised ridge predictor (fit on train folds) whose held-out correlation ρ_{oracle} is a *lower bound* on recoverability: $\rho_{\text{oracle}} > 0$ *proves* that this much of an un-

Table 1. Reconstruction from $\{I, II, V2\}$ on PTB-XL (fold 10). h (mV) is the deployable hallucination energy; ρ is the oracle correlation of the non-dipolar component with truth. The dipolar row’s $\rho=0$ is degenerate (it has no non-dipolar component); the generative row’s is genuine decorrelation.

seg	method	RMSE (mV)	h (mV)	ρ
QRS	dipolar	0.196	0.000	+0.00
	OLS	0.196	0.030	+0.13
	generative	0.228	0.097	+0.00
ST	dipolar	0.125	0.000	+0.00
	OLS	0.112	0.016	+0.28
	generative	0.149	0.063	−0.00
T	dipolar	0.140	0.000	+0.00
	OLS	0.105	0.028	+0.44
	generative	0.163	0.067	−0.00

observed lead’s non-dipolar residual is recoverable from the observed leads (a simple linear map attains it out of sample). For $S=\{I, II, V1, V3, V5\} \rightarrow \{V2, V4, V6\}$ (precordial interpolation) the non-dipolar residual is provably recoverable across the diagnostic segments (aggregate $\rho_{\text{oracle}} = +0.31$ over QRS/ST/T $\times \{V2, V4, V6\}$; +0.33 for QRS alone; Band A); the limb-6 control is lower but still nonzero ($\rho_{\text{oracle}} = +0.21$), which we use to check the deficit does not fire spuriously. We train a conditional 1-D DDPM (classifier-free guidance, RePaint measurement replacement) on NORM PTB-XL and score held-out NORM reconstructions with the certificate (Fig. 2; all quantities below are averaged over QRS/ST/T $\times \{V2, V4, V6\}$).

Two comparisons avoid a mean-vs-sample confound: the diffusion *posterior mean* (K -sample average, variance-free) and a single deployed draw. At the honest end ($w=1$) the posterior mean *matches* the oracle ($\Delta\rho = -0.00$): the model is competent and Band A is genuinely recoverable. As the guidance scale w rises the mean sharpens toward population-typical morphology and develops a *growing recoverability deficit* $\Delta\rho = \rho_{\text{oracle}} - \rho_{\text{mean}}$ — a distributional effect, not single-sample variance. The growth is monotone in the seed mean and in 3 of 4 independently trained models (all 4 for $w \geq 2$), and unambiguous at high realism: $\Delta\rho = +0.146 \pm 0.034$ at $w=6$ (4.3σ over seeds). Crucially, already in the operational regime ($w \leq 4$) — where RMSE is *flat* (0.214 mV, posterior mean) and the model stays faithful on the recoverable dipole subspace ($\rho_{\text{recoverable}} = +0.89$) — the deficit is present, if marginal ($+0.08 \pm 0.04$, 1.9σ): the fabrication is invisible to a scalar and localized only by the certificate. Only at extreme guidance ($w=6$) does the model visibly degrade on *every* metric (RMSE 0.317 mV, $\rho_{\text{recoverable}} \rightarrow +0.83$) — by then the fabrication is severe enough to move even RMSE, which the certificate had flagged all along. On the limb-6 control the linear reference is lower but nonzero ($+0.21$); the diffusion matches or exceeds it and

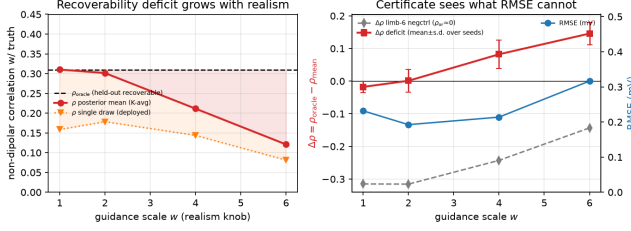


Fig. 2. Conditional DDPM, $S=\{I,II,V1,V3,V5\} \rightarrow \{V2,V4,V6\}$, held-out NORM PTB-XL. *Left:* the posterior mean tracks the held-out recoverable reference ρ_{oracle} at low guidance and pulls away as w rises (red band = a genuine recoverability deficit); a single deployed draw sits a fixed sampling-variance band below (orange). *Right:* the deficit $\Delta\rho$ (mean $\pm$ s.d. over 4 seeds) grows monotonically with the realism knob; RMSE stays flat through $w=4$ and rises only at extreme guidance, and the limb-6 control shows no positive deficit — the certificate sees, and localizes, what a scalar cannot until far later. All curves average QRS/ST/T $\times \{V2,V4,V6\}$.

the deficit stays negative ($\Delta\rho = -0.31$, QRS), so a *positive* $\Delta\rho$ marks a genuine failure to recover provable content, not the mere act of raising guidance.

4.5. Safety case: precordial ST from limb leads

The certificate warns *before* reconstruction that limb-6 $\rightarrow$ precordial is unsafe for an ST decision: $\kappa=6.7 \times 10^4$, effective rank 2, and the ST segment is only 0.74 dipolar, so precordial ST is largely Tier III. The danger is real, *bidirectional*, and RMSE-invisible: over 799 fold-10 records (Table 2), all methods have ~ 0.075 mV ST error, yet the generative reconstructor *fabricates* 263 phantom STEMI and the conservative OLS *masks* 360. Two honest caveats we report rather than hide: (a) even the dipolar reconstructor—blessed with $h \equiv 0$ —fabricates 226 STEMI, because that error lives *inside* the recoverable subspace (a contaminated dipole estimate) where h is blind; (b) the null-space flag catches only a minority of generative fabrications (18/263, held-out threshold). The dependable safety signal is therefore the *configuration-level* certificate: this lead set should not drive a precordial ST call, and κ says so with no reconstruction at all.

4.6. Guarantees under device shift

Calibrating on the Schiller CS device family and testing on the AT family (a genuine within-PTB-XL device shift), the ST/V4 interval keeps nominal 0.90 coverage in-distribution but drops to 0.82 on the shifted device—Tier II coverage is shift-sensitive. A small (300-record) recalibration on the new device restores it to 0.93 at a modest width increase (0.15 $\rightarrow$ 0.24 mV); naive likelihood-ratio weighting instead

Table 2. Limb-6 $\rightarrow$ precordial STEMI failures (PTB-XL fold 10, 799 records; ST at J+60 ms vs. PR baseline, 0.1 mV threshold). RMSE-invisible, bidirectional, and only partly flaggable—the reliable signal is the upfront κ warning.

method	fabricated	masked	ST err (mV)
dipolar ($h \equiv 0$)	226	165	0.074
OLS	6	360	0.075
generative	263	134	0.080

over-widens to trivial (1.00) coverage and is not a repair. Tier I exactness and the Tier III flag’s soundness are unaffected, as promised: only the Tier II *level* is shift-sensitive, and it is measurable and repairable.

Cross-dataset. On the geographically distinct Chapman-Shaoxing database [13] (349 Chinese hospital records, identical processing), the data-estimated dipole basis M_s transfers. Per-segment dipolarity matches PTB-XL (QRS 0.88/0.91, ST 0.76/0.72, T 0.76/0.74); the dominant QRS directions align (principal angles $\leq 21^\circ$); and the *well-conditioned* $\kappa_s(S)$ agrees (spanning $\{I,II,V1,V3,V5\}$: 2.3 vs 2.5). The lead-mixing operator T (Einthoven/Golberger) is dataset-independent by definition; the fit confirms the estimated M_s is too — save the lowest-variance dipole direction of the *less-dipolar* ST/T segments, and the limb-6 configuration, which is effectively rank-deficient in both cohorts ($\kappa_s \gtrsim 10^4$, its transverse dipole unobserved and its magnitude numerically unstable).

5. CONCLUSION

Reduced-lead ECG reconstruction is not just ill-posed but ill-posed with a *known physical structure*. Exploiting the cardiac dipole turns “trust the reconstruction” into a per-feature certificate: exact recovery where the dipole is spanned (with a conditioning $\kappa_s(S)$ that warns where it is not), a calibrated interval where the population constrains the residual, and a flag where nothing does. It tells you, *before any reconstruction*, which features a lead configuration can support—and shows that fabrication is a choice of *objective*, not an accident of the network. The certificate is reconstructor-agnostic and wraps any method.

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